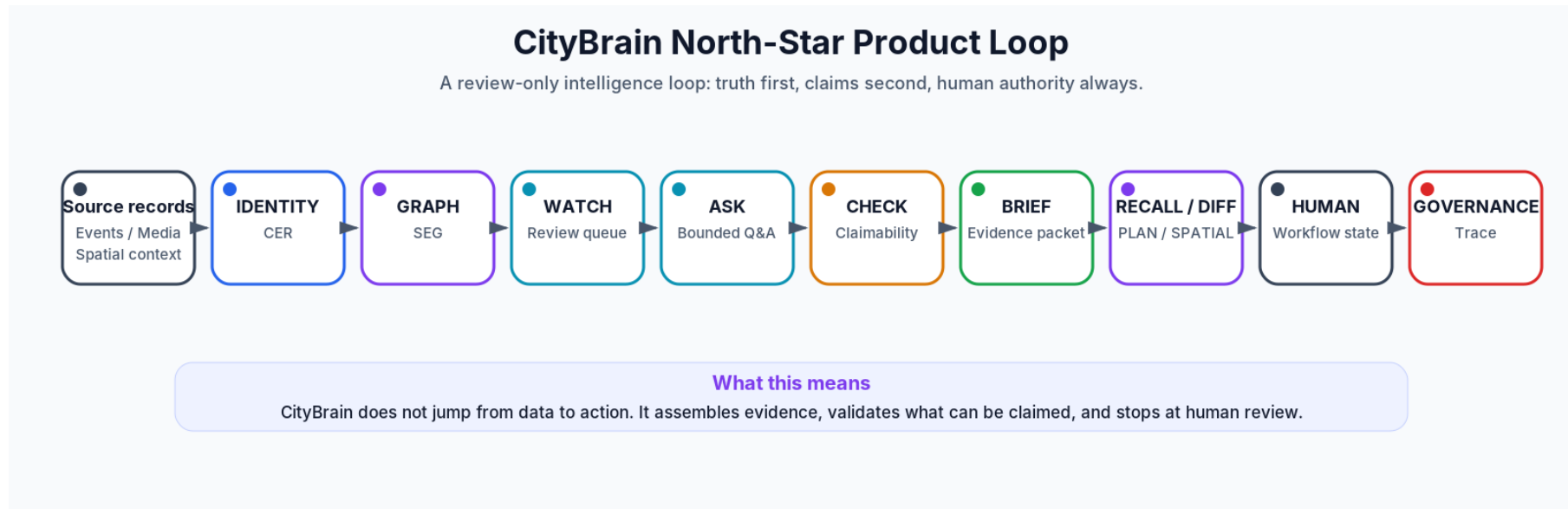


CityBrain Product Consolidation

Product picture, architecture, PM view, gaps, parked work and component plug-in map

Generated 2026-07-07 from Codex exports, conversation summaries and source docs



CityBrain is a governed review-only city intelligence cockpit that turns fragmented city records, events, media candidates and spatial context into evidence-backed human-review packets.

1. Executive picture

CityBrain is not a dashboard, chatbot, CCTV detector, graph demo or Omniverse scene. It is a governed city intelligence platform that helps people understand, validate, brief, review and eventually approve city workflows under explicit authority.

What can be safely said today

- Ingest and retain city/source records.
- Organize records into evidence-backed entities and context.
- Show source-backed records and review packets.
- Answer bounded ASK-style questions from sealed packets.
- Surface WATCH/review prompts in local/replay contexts.
- Generate briefs and evidence/context packets.
- Support review-only option sets and planning context.
- Connect candidate perception observations to review workflows.
- Show spatial/Omniverse/WebRTC proof paths.
- Run governed traces, ledgers, hash manifests and claim-boundary audits.
- Materialize labels and run one offline experimental forecast under explicit authority.

What cannot be safely claimed yet

- Production live operations.
- Public API / SLA.
- Certified citywide digital twin.
- Autonomous monitoring or dispatch.
- Enforcement/legal/certified finding.
- Official ticket/case/work-order creation.
- Product forecast surface.
- Live Watch consumption of a forecast model.
- Cross-city learned transfer.
- Trained ranking in operator surfaces.
- Production-grade execution adapters.

2. Product management view

Product Management Map: User Value by Audience

Operator

What needs review, what can I safely do next?

WATCH, CHECK, selected-item workspace, workflow state, safe next looks

Analyst / planner

What do we know, what changed, what is similar?

ASK, RECALL, DIFF, graph context, source-record 360

Executive / client

What is the story, value and risk boundary?

BRIEF, option-set summary, risk report, demo route

Data / twin engineer

Where is truth, geometry and identity?

Source registry, CER/SEG, spatial binding, data maturity

Governance / risk

What must not be claimed or executed?

CHECK, AuthorityEnvelope, no-action ledger, guardrails

ML / evaluation

What can be learned safely?

Outcome ledger, calibration, backtest harness, experimental forecast registry

Product modes and PM rules

Operator-facing modes

- WATCH: candidate review queue.
- ASK: bounded cited question answering.
- CHECK: evidence sufficiency / claimability validation.
- BRIEF: evidence packet generation.
- RECALL and DIFF: precedent memory and change review.
- PLAN / SCHEDULE / SPATIAL / WORKFLOW: review-only options, sequencing, spatial selection and local review state.

System and platform modes

- IDENTITY and GRAPH: entity resolution and relationship traversal.
- EVENT and PERCEPTION: event state and candidate observations.
- SIMULATE and OPTIMIZE: scenario modeling and constrained option generation.
- GOVERN: boundary, refusal and claim audit.
- QUALITY/MATURITY, FEDERATION, SYNTHETIC and PERSONA: commercial/platform expansion modes.

PM rule: the product unit is not a UI widget. It is a packet: canonical entity identity + graph context + evidence + CHECK report + authority envelope + workflow state.

3. Architecture blueprint

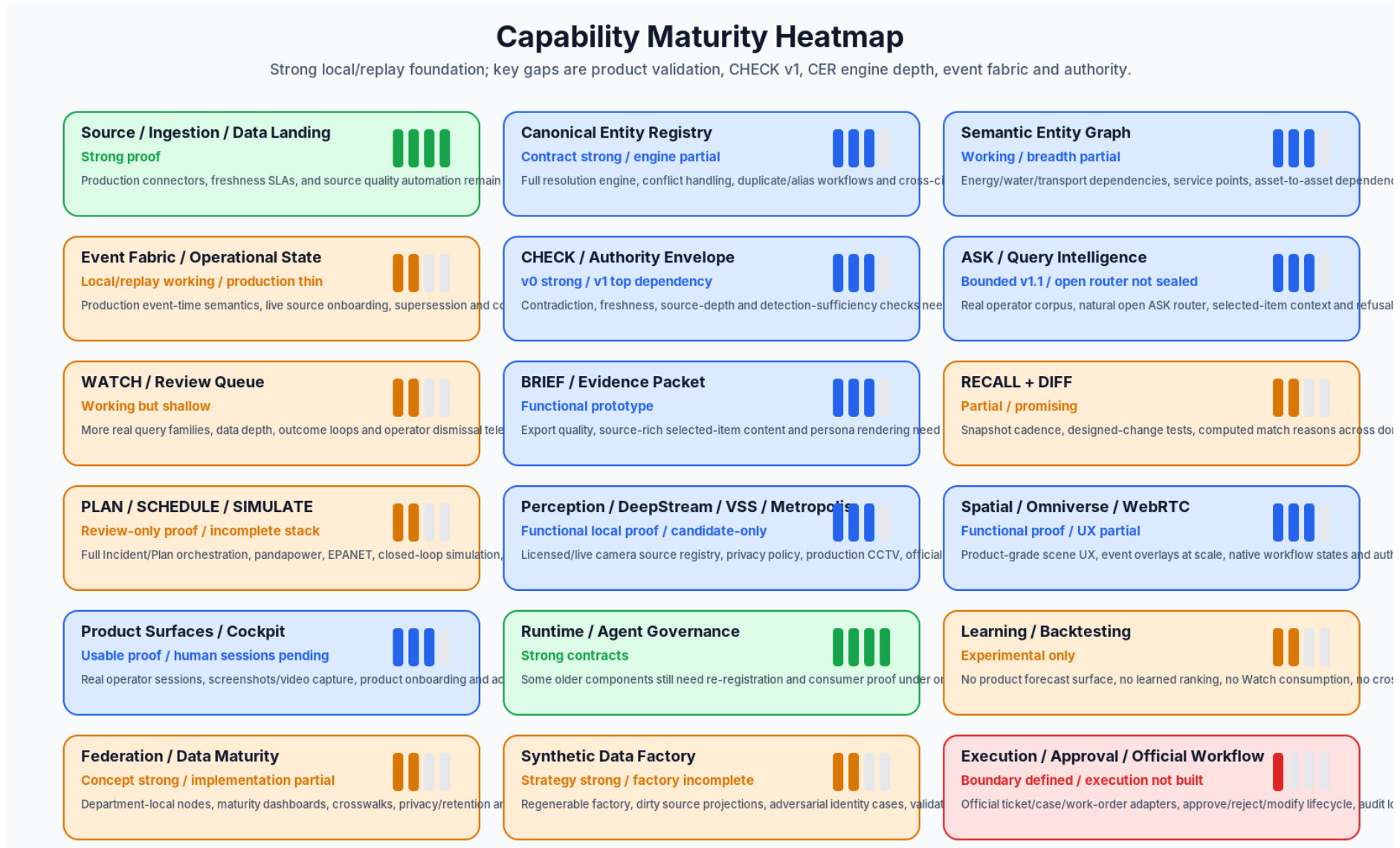


The hard architecture spine is: Canonical Entity Registry first, Semantic Entity Graph on top, Evidence/CHECK/Authority Envelope, review-only modes, human workflow state and governance trace.

Why CER before graph

The core problem is identity under ambiguity: mismatched IDs, inconsistent names, partial geometries, duplicates, stale records and unreliable source keys. The registry owns identity; the graph projects relationship intelligence only after identity contracts are stable.

4. Capability maturity map



Maturity summary table

Capability	Maturity	Main gap	Next move
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Source / Ingestion / Data Landing	Strong proof	Production connectors, freshness SLAs, and source quality automation remain incomplete.	Normalize ingestion manifests into one SourceRegistry v1.
Canonical Entity Registry	Contract strong / engine partial	Full resolution engine, conflict handling, duplicate/alias workflows and cross-city identity semantics are not complete.	Build CER Engine R1 with canonical_entities, source_entities, assertions, match_candidates and review queue.
Semantic Entity Graph	Working / breadth partial	Energy/water/transport dependencies, service points, asset-to-asset dependencies and temporal edge semantics remain thin.	Graph v2 over CER assertions with edge confidence, review_state and temporal validity.
Event Fabric / Operational State	Local/replay working / production thin	Production event-time semantics, live source onboarding, supersession and continuous monitoring not complete.	Event Fabric v1: append, replay, resolve, preserve, quarantine, materialize, query, trace.
CHECK / Authority Envelope	v0 strong / v1 top dependency	Contradiction, freshness, source-depth and detection-sufficiency checks need product-grade v1.	CHECK v1 with claim-to-evidence mapping and cannot-claim downgrade logic.
ASK / Query Intelligence	Bounded v1.1 / open router not sealed	Real operator corpus, natural open ASK router, selected-item context and refusal taxonomy still need validation.	Staged ASK router R1: boundary -> intent family -> target -> answer lens -> selected context.
WATCH / Review Queue	Working but shallow	More real query families, data depth, outcome loops and operator dismissal telemetry needed.	WATCH v1 family library tied to domain packs and CHECK v1.
BRIEF / Evidence Packet	Functional prototype	Export quality, source-rich selected-item content and persona rendering need hardening.	BRIEF v3 templates with operator/planner/executive/persona policies.
RECALL + DIFF	Partial / promising	Snapshot cadence, designed-change tests, computed match reasons across domains and precedent datasets need expansion.	Recall/DIFF R2 on real snapshots with source-level change classifier.
PLAN / SCHEDULE / SIMULATE	Review-only proof / incomplete stack	Full Incident/Plan orchestration, pandapower, EPANET, closed-loop simulation, uncertainty and calibration not complete.	One simulator end-to-end, starting with SUMO then pandapower/EPANET.
Perception / DeepStream / VSS / Metropolis	Functional local proof / candidate-only	Licensed/live camera source registry, privacy policy, production CCTV, official-	Perception candidate schema + source

		finding path not built.	registry + evidence clip retention policy.
Spatial / Omniverse / WebRTC	Functional proof / UX partial	Product-grade scene UX, event overlays at scale, native workflow states and authoring pipeline still need polish.	Hero-neighbourhood twin with event overlay and evidence card click-through.
Product Surfaces / Cockpit	Usable proof / human sessions pending	Real operator sessions, screenshots/video capture, product onboarding and accessibility not finished.	Epoch 4 review pilot: 5-10 real sessions, task success, confusion logs, export feedback.
Runtime / Agent Governance	Strong contracts	Some older components still need re-registration and consumer proof under one runtime contract.	Runtime recertification: no component closes without registered consumer and replay fixture.
Learning / Backtesting	Experimental only	No product forecast surface, no learned ranking, no Watch consumption, no cross-city learned transfer.	Keep offline until enough operator labels and CHECK calibration exist.
Federation / Data Maturity	Concept strong / implementation partial	Department-local nodes, maturity dashboards, crosswalks, privacy/retention and federation contracts need more depth.	Data Quality & Maturity R1 with identity fragmentation and source coverage dashboards.
Synthetic Data Factory	Strategy strong / factory incomplete	Regenerable factory, dirty source projections, adversarial identity cases, validation harness and replay scenario generation need formalization.	Synthetic Factory v1 around Dubai 217 community polygons if available.
Execution / Approval / Official Workflow	Boundary defined / execution not built	Official ticket/case/work-order adapters, approve/reject/modify lifecycle, audit log and monitoring after approval not productionized.	Approval Lifecycle R1 for draft/sandbox only; production execution remains future authority.

5. Gaps and parked work



Top gap register

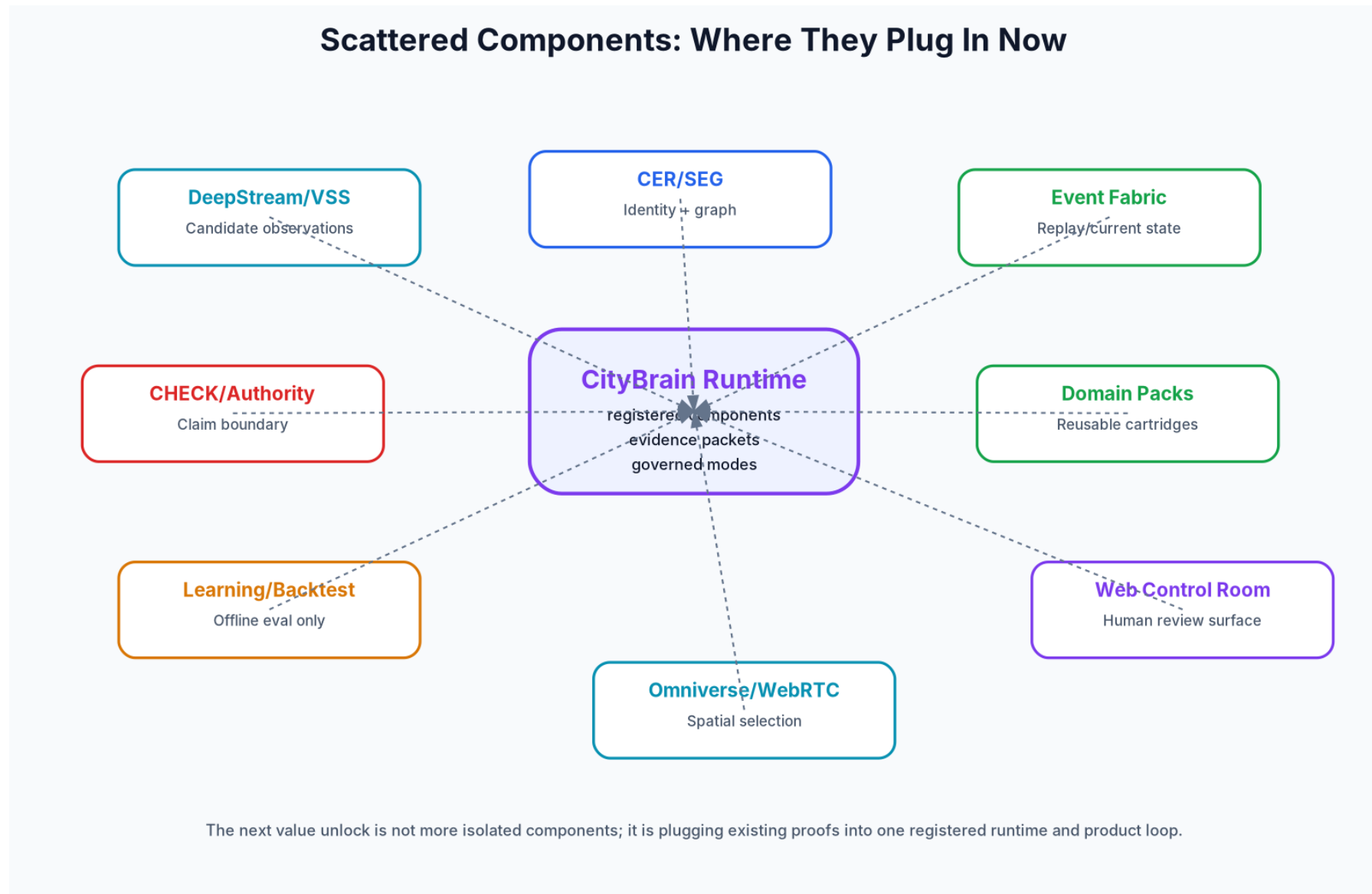
Priority	Gap	Why it matters	Parked / built	Next action
P0	CHECK v1 contradiction/source-	Claimability is the product trust	CHECK v0 exists but deeper rules	Build CHECK v1 with claim-to-

CityBrain Consolidation Pack - review-only product truth - generated 2026-07-07

	depth/freshness engine	layer; every mode depends on it.	remain open.	evidence mapping, stale-source checks, contradiction rules and cannot-claim downgrade.
P0	CER engine depth and conflict resolution	Graph, spatial selection and recall are unsafe if identity is unstable.	Contracts and examples exist; full engine is incomplete.	Build CER R1 and bridge all domain packs to canonical IDs.
P0	Real human operator sessions	The product surface is not truly validated without user behavior.	Several gates record human review/session capture pending.	Run 5-10 recorded review sessions, capture confusion, task success, dismissed WATCH items and export feedback.
P0	Event Fabric v1/v2 production depth	Without event fabric the cockpit is mostly static/replay.	Local/replay materialization exists; production live fabric is not done.	Local v1 first: append/replay/resolve/quarantine/materialize/query/trace; only later live onboarding.
P1	Execution authority / official workflow adapters	Clients will ask how review becomes action; today it must stay human-only.	HITL and proposal contracts exist, execution adapters are not built.	Define sandbox adapter only; no production execution until authority policy exists.
P1	Open ASK router and real operator corpus	ASK is the natural entry point to the product.	Staged router design exists; training/real operator validation not claimed.	Build router eval corpus with refusals, selected-item questions, queue questions and source-profile questions.
P1	Simulation stack beyond one review problem	PLAN/SCHEDULE/SIMULATE needs calibrated models to become more than option copy.	SUMO/cuOpt proof exists; pandapower/EPANET/closed-loop missing.	Pick one simulator end-to-end, starting with SUMO, then utilities.
P1	Synthetic Data Factory, not just packs	Spark testing and demos need reproducible worlds with known truth and deliberate messiness.	Gold/dirty/challenge/scenario strategy exists; factory not formalized.	Generate gold, dirty, challenge and scenario tiers with seeded reproducibility.
P1	Data quality / maturity cockpit	This is a strong commercial consulting wedge for departments.	Concepts exist; dashboards underbuilt.	Build maturity scorecards: identity fragmentation, source freshness, relationship coverage.
P2	Forecast productization	Offline experiment is promising but not usable by operators yet.	One frozen-replay forecast exists; no product surface or Watch consumption.	Keep offline until operator label depth and calibration gates pass.
P2	Cross-city learned transfer / dynamic investigation	Strategic differentiator but unsafe before labels and federation	Explicitly future-track / not armed.	Hold until Federation R1 and

		mature.		enough reviewed cases exist.
P2	Omniverse product UX and hero-neighbourhood twin	Spatial proof exists; clients need a legible demo body.	WebRTC/Kit proofs exist but UX polish remains separate.	Build one hero-neighbourhood twin, not a citywide twin first.

6. Component plug-in map

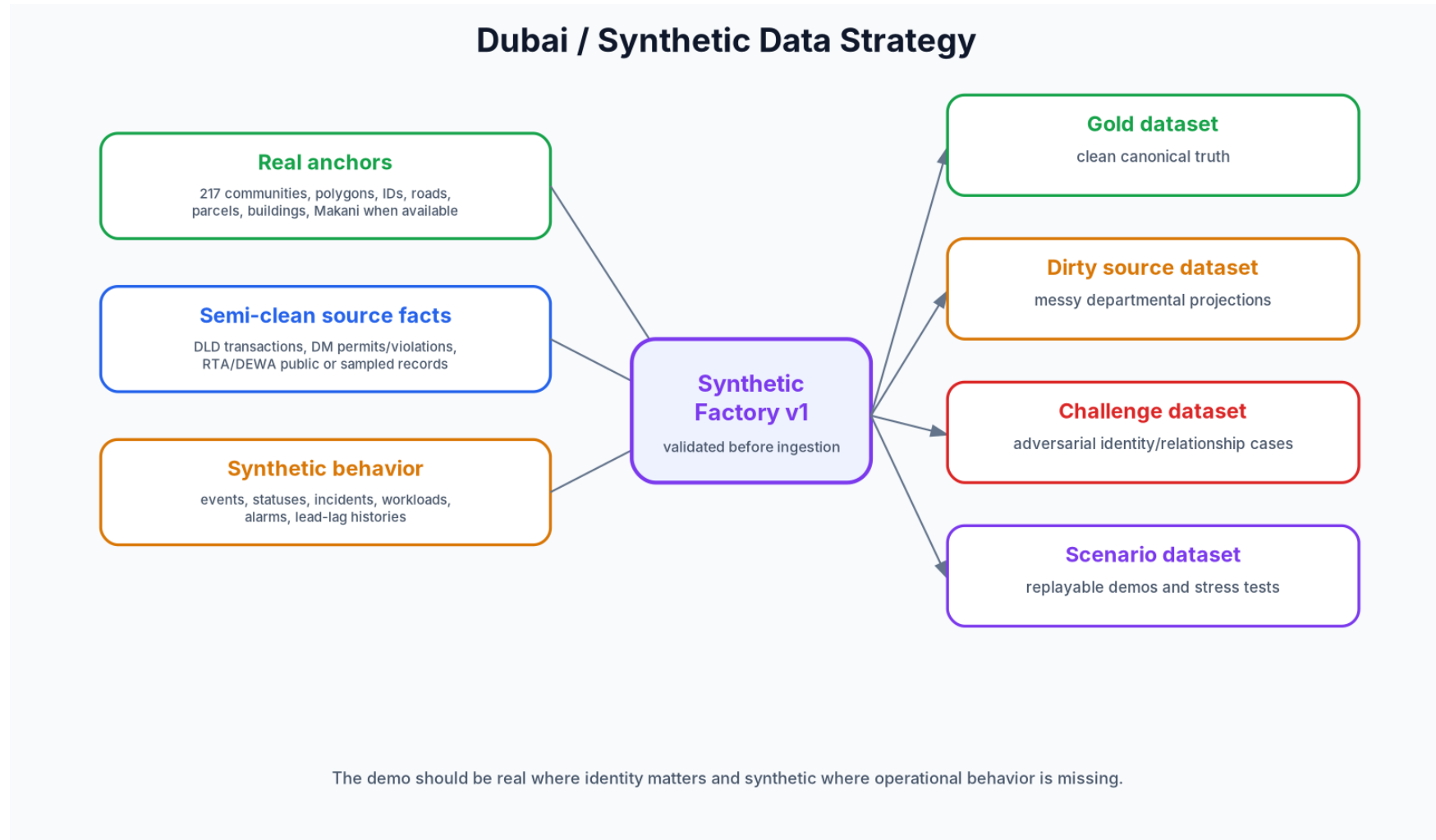


The next value unlock is not more isolated components. It is registering existing proofs into one runtime and product loop. Every component should have a component ID, version, owner, tool permissions, source classes, budget, consumer surfaces, replay fixture and release status.

Immediate plug-in opportunities

- DeepStream/VSS -> Event Fabric -> CHECK -> Review UI.
- CER/SEG -> Spatial binding so Omniverse selection is trustworthy.
- CHECK -> all modes, not just warning copy.
- Outcome ledger -> WATCH relevance learning before trained rankers.
- Data maturity -> client diagnostic product.
- Domain packs -> flows/cartridges instead of bespoke agents.

7. Data and synthetic strategy



The strongest Dubai strategy is real where identity matters, synthetic where behavior is missing. Real community IDs/polygons, Makani, roads/buildings and true counts should anchor the world; statuses, scenarios, events, workloads and stress histories can be synthetic and clearly labeled.

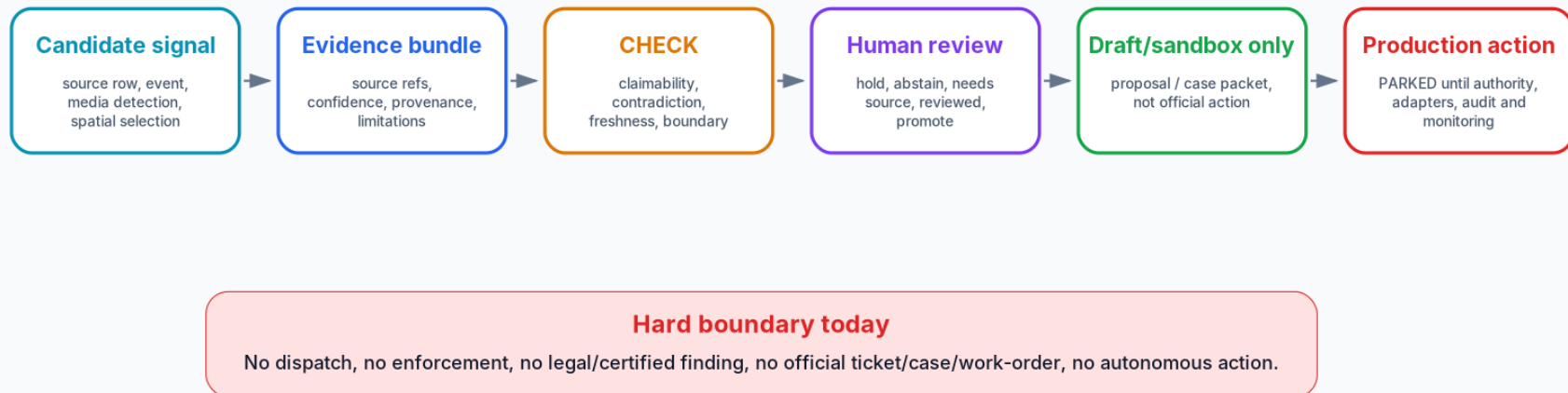
Four synthetic tiers

- Gold dataset: clean canonical truth.
- Dirty source dataset: messy departmental projections.

- Challenge dataset: adversarial identity and relationship cases.
- Scenario dataset: replayable demos and stress tests.

8. Governance boundary

Governance Boundary: Candidate Intelligence, Not Autonomous Action



The current product boundary is clear: candidate signals become evidence and review packets. They do not become dispatch, enforcement, legal findings, official cases or autonomous action.

9. Roadmap and next moves



Horizon	Goal	Deliverables
Now	Publish shared picture	Use this pack to align team/client language and stop definition-of-done drift.
0-2 weeks	P0 trust gates	CHECK v1 design, CER R1 design, human review pilot plan, Event Fabric v1 scope.
2-6 weeks	Reference spine R1	Event Fabric local v1 + CER engine + CHECK v1 + operator sessions + selected-item/BRIEF export.
6-10 weeks	Hero demo body	Hero-neighbourhood twin, event overlays, Watch family expansion, one simulator path.
10-13 weeks	Client pilot package	Data maturity dashboard, domain pack story, privacy/retention, pilot scorecard, client-ready demo script.

P0 decisions

- Publish this pack as the team/client product room.
- Build CHECK v1 and CER R1 before new intelligence.
- Run real operator sessions.
- Finish local Event Fabric v1.
- Build one hero-neighbourhood spatial demo with event overlays.

10. Scenario portfolio and demo script

Scenario / Flow Portfolio

Flow 1	Situational status Descriptive · Donor: NYC/Dubai · Cheap next cartridge; status of area X.
Flow 2	Construction compliance cascade Reactive / golden path · Donor: Dubai + NYC · Deep thread: vision -> graph -> evidence -> option/brief.
Flow 3	Resilient city: fire + road + air Reactive · Donor: London · Not yet proven as Flow 3; London foundation exists.
Flow 4	Crowd surge / major event Proactive · Donor: Melbourne/Singapore · Needs LTA/key and event/crowd data depth.
Flow 5	Flood / asset dependency cascade Proactive / forecast · Donor: NYC/London/ERA5 · Requires flood/weather/water/energy simulation.
Flow 6	Planned shutdown sequencing Proactive / domain swap · Donor: Oil & gas cartridge · Powerful story but separate domain pack.
Flow 7	Civic service + sensor fusion Reactive · Donor: Chicago · Strong flat-file/civic proof candidate.

Client demo arc

- The city boots: records, entities, graph and map/twin connect.
- A WATCH item appears: review prompt, not alert.
- The operator asks: bounded ASK answer from packet truth.
- CHECK validates: support, uncertainty, stale sources, cannot-claim.

- BRIEF packages: operator/executive/analyst view.
- SPATIAL grounds: same packet in map/Omniverse/WebRTC.
- Human boundary: hold, abstain, mark reviewed or draft/sandbox only.

11. Source crosswalk and evidence basis

This pack consolidates uploaded Codex exports, conversation summaries, CityBrain Evolution Digest, RTX Vision/Roadmap/SyntheticData docs, north-star product loop notes, VSS/Metropolis closeout, Barcelona and task-update files, and the previous consolidation master. It is a synthesis layer, not a forensic repo audit.

Recommended source-of-truth publishing layer

- Product Truth page.
- Architecture Spine page.
- Gap/Parked Register.
- ComponentRegistry export.
- Demo/Client Story pack.
- Monthly source crosswalk.